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Which inference is at risk? Assessment validity reasoning and generative AI

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ABSTRACT

Generative artificial intelligence has unsettled higher education assessment because it changes what student performances can reasonably be taken to mean. This conceptual article further develops a validity-centred account of this disruption, distinguishing several problems that are often collapsed in debate: construct underrepresentation, construct-irrelevant variance, attribution of performance, conditional extrapolation, and unsupported score use. It then proposes a practical reasoning sequence for assessment redesign that begins with specifying the intended claim and the AI conditions, proceeds through threat diagnosis and proportionate design response, and ends with local evidence. A compact Validity Reasoning Matrix illustrates how course and program teams can align concerns, threatened inferences, redesigns, and evidence needs. The contribution is both conceptual and practical: it offers a language for moving beyond detection and generic AI-proofing and a usable structure for defending assessment claims in AI-inclusive higher education.

KEYWORDS

Assessment validity; generative AI; higher education; argument-based validation

Introduction

Generative AI (GenAI) has moved from novelty to ordinary study infrastructure with unusual speed, and higher education assessment has become one of the places where this change is most visible. Recent student surveys suggest that AI use is already near-universal, but also unevenly distributed. Stephenson and Armstrong (2026), for example, reported that 95% of surveyed UK undergraduates had used AI in some form and that 94% had used GenAI for assessment-related work. Beckman et al. (2025) describe a GenAI divide among students, marked by differences in confidence, access, and educational support. At institutional level, McDonald et al. (2024) found extensive but uneven guidance across U.S. research universities, much of which leaves substantial interpretive work to instructors. Taken together, GenAI has categorically transformed the conditions under which student work is produced, interpreted, and used.

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The most visible response has been to treat GenAI as an academic integrity problem; if students outsource substantial parts of assessed work, grades may no longer support claims that they have met intended outcomes. Yet the assessment problem expands beyond mere rule violation, because the central concern is the meaning of the assessment evidence itself; in other words, it is a validity problem (Dawson et al. 2024). In a study embedded in a real university examination system, Scarfe et al. (2024) found that most AI-generated submissions went undetected and received competitive grades. Though some contemporary systems – notably Pangram – perform well under specific benchmark conditions, performance varies substantially across tools, task types, disciplines, model versions, and human–AI editing practices, and even favourable evaluations caution against treating detector scores as sole evidence in high-stakes decisions (Sun, Liao, and Ma 2026; Van Vlasselaer, Van Droogenbroeck, and Spruyt 2026). Detection-centred governance, or 'cop shit' in Moro's (2020) terms, is therefore an insufficient basis for upholding assessment validity: detector outputs are conditional, probabilistic signals that may prompt further inquiry but cannot by themselves establish misconduct or competence (Bassett et al. 2026).

Corbin, Bearman, et al. (2025) recently described AI and assessment as a wicked problem, where security, learning, fairness, authenticity, and workload cannot be optimised through a single, stable solution. Consequently, some scholars advocate for structural change rather than unenforceable declarations or instructions (Corbin, Dawson, and Liu 2025). The limits of apparently integrity-preserving assessment design also predate GenAI: Ellis et al. (2020) found that authentic assessment tasks were routinely outsourced through contract cheating. Kofinas, Tsay, and Pike (2025) show that the same caution remains necessary under GenAI, while Arnold and Croxford (2025) question whether the language of authenticity obscures as much as it clarifies. Chirikov, Smirnov, and Kizilcec (2026), in turn, argue that GenAI use and misuse call for assessment reform in higher education rather than piecemeal responses to misconduct. Across these contributions, a shared issue comes into view: GenAI forces higher education to ask more carefully what its assessments are supposed to evidence.

This article develops that issue through the concept of assessment validity. In the Standards for Educational and Psychological Testing, validity refers to the degree to which evidence and theory support interpretations of scores for proposed uses (American Educational Research Association, American Psychological Association, and National Council on Measurement in Education 2014). Validity is nevertheless a contested concept, with accounts variously foregrounding or bracketing different dimensions of this idea; for example, whether consequences form a part of validity itself or are a related evaluative concern (Newton and Shaw 2014). This article adopts an argument-based interpretation-and-use account (Kane 2006, 2013) because GenAI unsettles the inferential steps connecting an observed performance to a claim about a student and subsequent decisions. Here, *construct* refers to the student capability an assessment is intended to represent, such as writing fluency or AI-supported professional judgement. It is not synonymous with the task used to elicit that capability. The term *score* is used broadly to include marks, grades, pass-fail judgements and other evaluative outcomes. This perspective asks whether an

assessment supports a specified claim and use under stated conditions, rather than whether a particular format is valid in general.

Dawson et al. (2024) reframed cheating as but one threat to assessment validity, placing the emphasis instead on whether the assessment evidence supports claims about students' capabilities. This article addresses follow-up questions emerging from this validity frame: how can threats to validity be diagnosed and differentiated, and what can be done about it? The article makes three contributions. First, it reframes AI-related assessment concerns as disruptions to *specific* validity inferences. Second, it clarifies distinctions between AI use and construct mismatch, task alignment and score contamination, authorship and ownership of meaning, assessment and target contexts, and supported interpretations and justified decisions. Third, it offers a compact reasoning sequence and matrix for course and program teams.

Why AI and assessment is not one problem

Consider a familiar case. A student submits an excellent take-home essay in a course where GenAI use is permitted or, at least, difficult to prevent. Later, in a supervised oral explanation of the same ideas, the student performs poorly. The immediate interpretation may be that the earlier assessment has been compromised. The discrepancy, however, can point to several different problems, and each problem calls for a different kind of redesign. Table 1 shows how the same discrepancy can admit different explanations and raise different interpretive questions. Poor oral performance alone does not establish whether, or how, the essay assessment was compromised.

Validity as claim, conditions, and warrant

Validity is not a property of an instrument in isolation but the degree to which evidence and theory support interpretations of assessment outcomes for intended uses, in a particular context (American Educational Research Association, American Psychological Association, and National Council on Measurement in Education 2014). In Kane's argument-based validity, validation begins by specifying an interpretation/use argument – the chain of claims linking observed performance to score meaning and decision use – and then evaluating that argument through *warrants* and *backing* for each link (Kane 2006, 2013). A warrant justifies why, under stated conditions,

Table 1. One assessment discrepancy with multiple possible validity explanations.

Possible explanation of the discrepancy	Question raised for interpretation
GenAI enabled the essay to be completed without the intended reasoning, synthesis, or disciplinary judgement being fully enacted.	What capability does the essay now provide evidence of?
The essay score partly reflects differences in tool access, prompting experience, model quality, or marker response.	How much of the score reflects the intended student capability rather than AI-related conditions?
The product is strong, but the student cannot explain, defend, or reuse its ideas.	What aspects of the performance can reasonably be attributed to the student?
The essay and oral follow-up elicit different capabilities under different conditions.	Were they intended to support the same claim, and should their results be expected to agree?
Both performances provide relevant evidence, but it is unclear how they should be weighted or combined.	What decision can the combined evidence justify?

the data count as evidence for a particular claim; backing consists of the theory and empirical evidence supporting that justification. For independent disciplinary reasoning, the warrant is that the quality of an argument reflects the student's evaluation of evidence and construction of a justification. Backing here might draw on a cognitive account of those processes and empirical studies examining whether they explain students' assessed performances. In other words, backing supports the connection from an observable performance or product to student capability.

A common extension of argument-based validation distinguishes six linked inferences (Chapelle, Enright, and Jamieson 2008; Chapelle 2020; Figure 1). *Domain* asks whether tasks adequately represent the target domain and elicit relevant performance; *scoring,* whether observed performance is translated into a score accurately and consistently; *generalisation*, whether the score would remain sufficiently stable across relevant tasks, raters, occasions, and assessment conditions; *explanation*, whether score differences can be attributed to the targeted student capability rather than alternative causes; *extrapolation,* whether that interpretation extends to performance in target contexts beyond the assessment event; and *utilisation*, whether it provides adequate grounds for the proposed decision or action.

These inferences form a chain because each takes the claim supported by the preceding link as its premise: weakness upstream restricts what can be defended downstream. Design reasoning also runs in reverse, because the intended target context and use determine what task sampling, scoring consistency, and attribution evidence are required upstream. Messick's unified view complements this structure by emphasising multiple forms of validity evidence and threats such as construct underrepresentation and construct-irrelevant variance (Messick 1995).

GenAI enters this chain by altering performance conditions, that is, the conditions under which students generate the performance from which the evidence is drawn. The term *AI conditions* (Figure 1) refers specifically to GenAI-related dimensions of broader conditions, including tool access, permitted functions, model versions, disclosure expectations, verification opportunities, equity of support, and data or privacy constraints. These dimensions are foregrounded because GenAI can alter which capabilities students must exercise to produce an assessed performance. Other conditions, including time limits, collaboration, and accessibility arrangements, remain part of the same validity argument.

If one student has a premium model, a second receives detailed staff coaching, and a third treats AI use as prohibited because the policy is unclear, these differences shape the possible interpretations and uses of assessment scores. Applying the

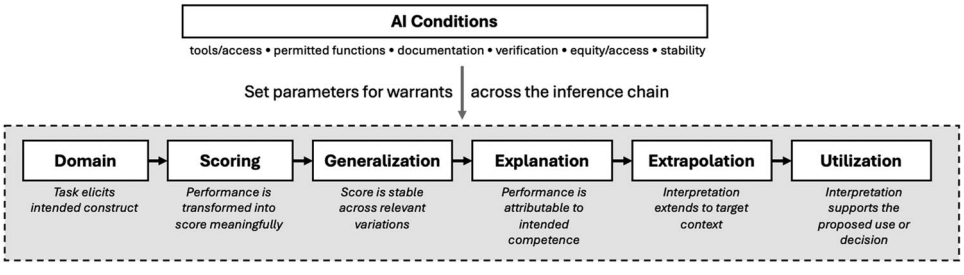


Figure 1. AI conditions as part of the validity argument.

framework therefore involves deciding what capability students must demonstrate and what role AI may play, then aligning access, support, and permitted use with that construct. For example, where differences in tool access are irrelevant to the intended capability, providing comparable access can reduce an unintended source of score variation. Specifying these arrangements and checking their implementation gives the validity argument a defined basis: it clarifies the conditions under which interpretations require justification and when changed conditions warrant reconsideration.

AI-inclusive assessment is used here as a provisional umbrella term for settings where GenAI is permitted, integrated into the task, or nominally prohibited without reliable evidence of compliance. Existing classifications distinguish AI-free, AI-assisted, and AI-integrated assessment, program-level lanes, and levels of permitted use (Bridgeman, Liu, and Weeks 2024; Digital Education Council 2025; Perkins et al. 2024). Their shared value is to clarify various relevant performance conditions. One argument developed in this article is that umbrella terms like *AI-inclusive assessment* must give way to specific arrangements for access, permitted functions, disclosure, verification, and tool stability, and to the validity argument those arrangements require (see Appendix A for an exemplary AI conditions statement template).

Diagnostic distinctions for AI-inclusive assessment validity

The following distinctions locate AI-related pressures within a validity argument and separate problems often collapsed under broad labels such as cheating or academic integrity. These selective distinctions locate assessment vulnerabilities that GenAI may amplify or reconfigure (see Appendix B for a broader overview). Diagnosis identifies the primary location and nature of a problem without implying mutually exclusive categories. Consequences and incentives are considered separately as cross-cutting concerns.

AI use can lead to construct mismatch

A first distinction concerns the difference between AI use as a behaviour and construct mismatch as a validity problem. In validity theory, the key question is whether the assessment elicits evidence of the construct that the score is intended to represent. The Standards describe construct underrepresentation as a failure to capture important aspects of the intended construct and construct-irrelevant variance as score variation caused by factors outside that construct (American Educational Research Association, American Psychological Association, and National Council on Measurement in Education 2014). GenAI use is therefore not automatically a validity threat. If the intended construct is AI-supported professional judgement, appropriate GenAI use may be germane to the construct; if the intended construct is unaided disciplinary reasoning, independent writing fluency, or unsupported problem solving, the very same AI use means that students are bypassing the target performance.

GenAI makes this established distinction newly salient. Borges et al. (2024) found that GPT-4 could answer a substantial proportion of university-level STEM assessment questions, showing how readily visible task requirements may be met without

the intended human performance being fully enacted. Xia et al. (2024), in their scoping review, likewise show that GenAI is reshaping how assessment tasks function across higher education. This means that a task that once supported a claim about independent problem solving may now support a different claim about tool-supported evaluation, verification, synthesis, or judgement, highlighting the insufficiency of ascribing validity to assessment types, as opposed to specifying construct and AI conditions together for each new assessment.

A task can target the intended construct while scores reflect the wrong variance

A second distinction separates construct representation from score contamination. An assessment may still be aligned with an intended construct, while the scores partly reflect unintended sources of variance. Construct-irrelevant influences on marking predate GenAI. For example, experimental research has shown that subjective grades can be affected by information about students unrelated to the quality of their work (Malouff and Thorsteinsson 2016). GenAI introduces additional variance sources. Two students with comparable disciplinary understanding may receive different scores because one has a better model, more prompting experience, stronger AI literacy, or more time to iterate. These skills may be educationally important or even part of the construct; where they are not, they enter the score as noise. Evidence of uneven access, confidence, and support in students' GenAI use makes this distinction consequential (Beckman et al. 2025).

GenAI also gives markers a new potentially biasing cue *via* disclosed or suspected AI use. Luo and Dawson (2026) show that teachers may make value judgements about GenAI-assisted work, including whether it should be marked down. If such judgements affect grades independently of stated criteria, they reproduce and potentially amplify an established scoring problem where scores no longer mean what the rubric says they mean. Detector-based responses can compound the issue because classification performance varies systematically across tasks and disciplines, with formulaic STEM writing particularly susceptible to algorithmic bias (Sun, Liao, and Ma 2026). Attempts to restore assessment security through detection may therefore introduce new construct-irrelevant variance and threaten fairness and score interpretation.

Authorship is not the same as ownership of meaning

A third distinction concerns the explanation inference: whether observed performance can be attributed to the intended student competence. Integrity debates often frame this as an authorship question – who produced the words, code, image, or solution? Validity reasoning leads to different questions: what claim about the student does this performance warrant? While outsourcing of student work naturally long predates GenAI (see e.g. Bretag et al. 2019), GenAI intensifies the problem by making its boundary less discrete: assistance is immediate, iterative, and often interwoven with students' own drafting rather than outsourced wholesale. A student may use AI for drafting, translation, or rephrasing while still exercising substantive judgement,

verification, and conceptual control. Conversely, a student may submit a sound product without being able to explain the reasoning, justify the choices, or use the ideas in a related context (Walton et al. 2026).

Ebrahimzadeh, Shibani, and Buckingham Shum (2026) develop this idea as coauthorship integrity: students remain accountable for the AI-assisted work they submit. The warrant connecting an AI-assisted product to student competence requires backing: product quality alone may not establish that the student exercised the targeted understanding or judgement (Scarfe et al. 2024). GenAI therefore widens an established evidentiary gap, making complementary evidence of students' contribution and understanding particularly important. Operationally, this might mean gathering complementary evidence that the student exercised the targeted understanding and judgement instead of policing provenance. For an AI-assisted case analysis, a student might be asked to justify a recommendation, explain how the cited evidence supports it, and identify a limitation. An annotated decision or focused follow-up can thus provide evidence of conceptual control of the submitted work. Such measures should also be proportionate. For example, not every essay needs a complementary oral exam, as oral assessments are themselves multidimensional and elicit performance under different conditions (Joughin 1998).

Evidence under one set of conditions may not support claims under another

A fourth distinction concerns extrapolation. Even when a score is valid under one set of conditions, it may not support claims under another. A score from AI-supported coursework may validly indicate AI-supported disciplinary performance, without also indicating independent competence in an AI-restricted context. Conversely, an AI-restricted examination may underrepresent the competence required in professional settings where responsible AI use is routine. Boud and Falchikov (2006) made a related point before GenAI: assessment should be aligned with the forms of capability students need beyond the course. GenAI can both amplify context mismatch by making performance conditions more variable and alter the target context itself when responsible AI use becomes part of professional capability. Therefore, where scores are intended to support a common interpretation, the claim and assessment conditions should be aligned in advance. Moreover, extending that interpretation to different target conditions requires evidence that it continues to hold.

This distinction matters for program-level decisions, certification, and public trust. Programs may need AI-supported tasks that capture contemporary professional practice alongside differently constrained tasks sampling independent competence. The central extrapolation question is whether performance under the assessment conditions supports claims about what the student can do in the intended target context.

A defensible score interpretation does not justify every use

A fifth distinction concerns whether a supported interpretation justifies a particular decision or action. Whereas preceding inferences address what performance means and whether that interpretation extends to a target context, utilisation concerns its

adequacy for the proposed use (Bachman 2005). A defensible, bounded claim may still be insufficient for progression, certification, or selection.

An AI-integrated capstone may credibly demonstrate AI-supported professional judgement under relevant conditions. Further warrants are nevertheless needed to justify progression thresholds, the capstone's weighting against other evidence, and the resolution of inconsistent results. Similarly, an oral follow-up may provide evidence of understanding, but using hesitation or vague responses to nullify an otherwise well-supported grade requires separate justification. In both cases, relevant evidence may be insufficient or inappropriately weighted for the decision.

Evidence for utilisation therefore concerns decision thresholds and stakes, the sufficiency and combination of evidence, and safeguards such as moderation, review, and appeal. Programmatic approaches accordingly distinguish individual assessment data points from high-stakes decisions, grounding the latter in multiple, longitudinal sources of evidence (Baartman and Quinlan 2024). GenAI-focused guidance similarly recommends multiple, contextualised approaches and targeted assurance at key points of progression and completion (Lodge et al. 2025). Upstream validity is necessary, but does not by itself justify every use.

Additional consideration: consequences and incentives

The inference chain in Figure 1 identifies the warrants needed to interpret and use assessment evidence. Assessment regimes have long shaped what students attend to Gibbs and Simpson (2005), which strategies appear rational, and how burdens and opportunities are distributed. Whether such consequences form part of validity proper remains contested (Cizek 2016; Messick 1995). They are therefore treated here neither as an additional inference nor as a synonym for utilisation, but as a cross-cutting design consideration. Consequences can bear on whether a proposed use is defensible, but they can also feed back into earlier inferences by changing performance conditions, participation, rater behaviour, or access. Thus, when an AI-compatible task rewards polished products obtainable through delegation more than defensible disciplinary reasoning, the problem is not only one of student conduct but also one of the incentive structure created by the assessment regime (Fawns and Schuwirth 2024). These effects are revisited below as trade-offs of assessment redesign.

Validity reasoning and assessment redesign

Redesigns create trade-offs. In response to GenAI, institutions have considered oral exams, invigilated assessments, authentic tasks, AI bans, AI declarations, detector use, standardised tool access, and process documentation. But as Dawson et al. (2024) have pointed out, redesign moves may strengthen one inference while weakening another. For example, oral defences may improve attribution but reduce reliability or accessibility. Authentic tasks may improve extrapolation but make scoring more complex and, as Kofinas, Tsay, and Pike (2025) show, do not remove integrity concerns by default. Corbin, Dawson, and Liu (2025) convincingly argued that structural changes are needed. But these, too, redistribute validity pressures

rather than eliminating them. Moreover, weaknesses at one point in the inference chain can propagate downstream; for example, reliable scoring cannot rescue a task that underrepresents the intended construct, or a well-supported interpretation does not by itself justify a high-stakes decision. Within a complex assessment regime, this does not mean that every task must support every inference equally. An AI-supported project might contribute domain and extrapolation evidence about contemporary professional practice, while a supervised follow-up contributes complementary explanation evidence about the student's understanding. The utilisation inference then concerns whether the rule for combining these sources is adequate for the intended decision.

The proposed redesign sequence has five steps (see Figure 2). First, specify the claim: what does a high score mean? Second, specify the AI conditions under which that interpretation is meant to hold (see Appendix A for an example). Third, locate the vulnerable inference by asking which warrant the concern most directly calls into question, using the distinctions above and the examples in Table 2 and Appendix B. Fourth, choose a design response that addresses the diagnosed pressure while checking what it changes elsewhere in the chain. Fifth, identify what local evidence is sufficient for the stakes. Fawns, Boud, and Dawson (2026) provide a useful bridge through their framework for practical assessment validation, which distinguishes evidence from product, process, performance, and practice.

In the proposed redesign sequence, proportionality should be carefully considered. The scope of local validation should reflect the assessment's stakes. Low-stakes formative tasks may require only clear condition statements, student-facing exemplars, and modest monitoring. High-stakes progression, award, or certification decisions require stronger evidence: complementary secure tasks, moderation, explicit mapping between outcomes and assessment mix, and review of equity effects. Williamson (2026), although focused on marking, usefully emphasises transparency, fairness, and accountability when AI enters high-stakes assessment systems. Those principles align with validity reasoning when they are attached to specific claims and uses rather than left as general assurance language.

Table 2 illustrates how a concern can be linked to an inference under pressure, a design question, and relevant evidence. At course level, it can help an instructor decide whether an assessment task needs redesign, additional attribution evidence, or clearer AI conditions. At program level, it can reveal whether a curriculum relies

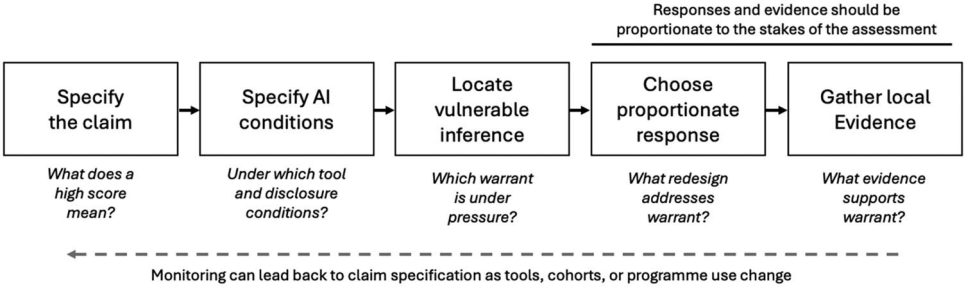


Figure 2. Validity-centred redesign sequence.

Table 2. Validity reasoning matrix.

If the concern is…	Primary inference pressure	The design question becomes…	Proportionate evidence might include…
Students can produce a strong product while bypassing target reasoning.	Domain	Does the task still elicit the intended construct under the stated AI conditions?	Task–outcome mapping; samples of students' reasoning
Scores reflect unequal tool access, prompting skill, or marker response.	Scoring	Do scores mainly reflect intended competence and criteria rather than AI-related differences?	Information on tool access; marked work checked against criteria
Student results change across comparable tasks, occasions, or other relevant conditions.	Generalisation	Would the score interpretation hold across relevant samples of performance and conditions?	Results across comparable tasks; results across occasions
A strong product may not reflect student understanding.	Explanation	Can the student account for and defend the submitted work?	Students' explanations of key decisions; supervised follow-up work
AI-supported performance is used to infer competence under different conditions.	Extrapolation	Does the performance support claims about capability in target contexts?	Performance in target settings; placement or supervisor feedback
A supported interpretation is given too much weight or translated into an unsupported decision.	Utilisation	Do the evidence and decision rule justify the proposed action at these stakes?	Rationale for thresholds and weighting; records of moderation
The regime creates harmful incentives, inequities, or other unintended effects.	Consequences*	What effects does the regime create, for whom, and do they warrant revision?	Student and staff feedback; patterns in participation and outcomes

Note. *Consequences are considered cross-cutting concerns, not a distinct inference.

too heavily on one kind of evidence, such as polished take-home products, while lacking secure or performance-based evidence for high-stakes claims. The matrix also shows why policy statements alone are weak evidence. They may specify rules, but they do not show whether the resulting scores support the intended interpretations.

A useful distinction is between documentation, evaluation, and monitoring evidence. Documentation asks whether the stated conditions are actually in place: students know what is permitted, support is available, and scoring criteria match the claim. Evaluation asks whether a mitigation strengthens the warrant it targets, for example whether an oral defense actually distinguishes students who can explain their work from those who cannot. Monitoring asks whether the argument remains plausible as cohorts, tools, policies, and teaching practices change.

Worked example: one essay, different claims

Consider again a final-year take-home case-analysis essay. Students must diagnose an institutional problem, use course concepts to interpret the case, and justify a recommended course of action. Before GenAI, a strong essay might have been treated as reasonable evidence of independent disciplinary reasoning, writing ability, and readiness for later professional judgement. Under AI-inclusive conditions, however, that compound interpretation is no longer safe by default. The question is

which of those claims still needs to be made, under which conditions, and for what use. Table 3 shows why disagreements about making an essay AI-safe may reflect different intended claims: independent reasoning, AI-supported professional practice, writing fluency, or readiness for progression. These require different conditions and evidence, while purely formative use may justify a more modest evidentiary demand.

Discussion: what a validity-centred account adds

The framework makes assessment redesign more explicit by connecting the intended claim, the vulnerable inference, and the evidence needed to support it. Its wider implications concern authenticity, constructive alignment, and the combination of evidence at program level.

A validity-centred account also offers a way to reframe authenticity in assessment. Authentic assessment has long been attractive because it points to a genuine limitation of many conventional formats; they can be detached from the complex, situated, and consequential uses of knowledge that higher education is meant to prepare students for (Gulikers, Bastiaens, and Kirschner 2004; Villarroel et al. 2018; Wiggins 1990). From a validity-centred perspective, however, authenticity is neither a competing principle alongside validity, nor a feature that protects a task from AI use (Fawns et al. 2025; Kofinas, Tsay, and Pike 2025). Instead, authenticity can provide backing for specific validity inferences, especially domain representation and extrapolation. In this view, assessment is authentic to the extent that it samples relevant features of the target practice, under defensible performance conditions, in a way that supports the intended interpretation and use of the score. Thus, authenticity becomes a claim that needs warranting, not a label that settles the warrant. A task may be authentic in its scenario but weak in attribution evidence or authentic to current professional practice but misaligned with educational aims. The framework therefore helps identify which validity inferences authenticity-oriented design choices may strengthen or weaken, and what evidence is needed to support the intended assessment claims.

Table 3. Different claims and uses for the same case-analysis essay.

Intended claim or use	Possible assessment approach	Relevant evidence
Independent disciplinary reasoning	Restrict AI for the reasoning being assessed; supplement the essay with a short supervised analytic task.	Reasoning demonstrated in the supervised task.
AI-supported professional analysis	Permit AI; assess verification, source use, and justification of key decisions.	Annotated decisions; explanations of source use.
Writing fluency	Restrict generative language support; assess writing separately from disciplinary content.	A writing sample produced under the specified conditions.
Capability in target academic or professional settings	Combine tasks that reflect the AI access and constraints of the target settings.	Performance in those settings; placement or supervisor feedback.
Formative use to guide learning	Use the essay provisionally to identify learning needs; seek follow-up evidence before making stronger claims.	Student explanations; subsequent performance on a related task.

The same logic also connects validity reasoning to constructive alignment. The claim that initiates the validity argument should be embedded in a wider learning design in which intended learning outcomes, teaching and learning activities, assessment conditions, and criteria cohere (Biggs 1996; Biggs and Tang 2011). The advent of GenAI can change not only how assessment evidence is produced, but also what kind of capability the course or program is trying to develop. If responsible AI-supported performance is part of the intended learning outcomes, then students need opportunities to learn, practice, and receive feedback on the relevant forms of judgement, verification, critique, and accountable tool use, and the assessment criteria must make those expectations visible. In this sense, AI-related capability, including what is often discussed as AI literacy, does not solve the validity problem from outside the assessment system. It has to be located within the alignment between learning goals, learning activities, assessment conditions, and, ultimately, score interpretation and use.

Individual task redesign cannot alone secure program-level claims. These depend on how evidence is combined across the curriculum, including the product, process, performance, and practice sources distinguished by Fawns, Boud, and Dawson (2026). A program dominated by polished take-home products may lack sufficient attribution evidence, while exclusive reliance on supervised AI-free assessment may underrepresent contemporary professional practice. The framework helps program teams examine whether their assessment mix supports the capabilities and decisions at stake.

Limitations and future research

The argument offered here is conceptual and design oriented. The suggested validity reasoning approach is not a validated protocol for assessment redesign, and its practical value should therefore be examined empirically. Future studies could investigate whether educators and program teams can use the matrix consistently, whether it supports more robust redesign decisions, and what kinds of local evidence are feasible under ordinary workload constraints. Student interpretation also matters: AI condition statements may reduce ambiguity, but they may also be misunderstood, unevenly enacted, or experienced as shifting responsibility onto students.

The framework can structure disciplinary deliberation but cannot determine whether AI-supported performance should be central, peripheral, or restricted in a particular field. Comparative work across disciplines could draw on established distinctions between performance achieved with technology and learning demonstrated without it (Salomon, Perkins, and Globerson 1991), alongside sociomaterial accounts of capability arising through relations among people, artefacts, and settings (Fenwick 2010). Where assessment claims involve embodied skill, interpersonal judgement, ethical responsibility, or creative production these perspectives raise interesting questions about when material resources are integral to the capability being assessed and what evidence of individual capability is also required.

Finally, future work should examine the consequences of redesign. Measures introduced in response to GenAI may improve attribution or security while affecting student anxiety, accessibility, workload, equity, trust, and learning strategies. Messick's (1995) consequential view of validity and the fairness standards in American

Educational Research Association, American Psychological Association, and National Council on Measurement in Education (2014) suggest that these effects are not peripheral. If an assessment regime becomes more secure but less equitable, less educationally productive, or less aligned with long-term learning, these changes may weaken the justification for particular interpretations or uses, or provide broader educational grounds for redesign.

Conclusion

Higher education assessment often relies on tacit continuity between task, performance, product, grade, and claim. GenAI therefore exposes something that has been true all along: validity depends on explicit claims, explicit conditions, and defensible warrants. When AI changes how performances are produced, it changes what those performances can be taken to mean. Assessment redesign therefore requires institutions to specify the claims they intend to make, the AI conditions under which those claims should hold, and the evidence needed to support them. The framework connects these questions to particular inferences and possible design responses. Its contribution is to distinguish problems with similar symptoms and make the assumptions behind assessment decisions explicit and open to scrutiny.

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Code availability

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Use of generative artificial intelligence

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Appendix A. Exemplary AI conditions statement template

Table A1. AI conditions statement template.

Field	Content to specify
Assessment regime	AI-free, AI-assisted, AI-permitted, or AI-required.
Stakes	Formative, low-stakes summative, high-stakes summative, progression, award, or certification.
Approved tools and access	Named tools, institutional or student-provided access, device/connectivity assumptions.
Permitted functions	Brainstorming, outlining, drafting, editing, translation, coding, data analysis, source support, or other.
Non-permitted functions	Direct answer generation for specified parts, hidden automation, fabricated references, undisclosed outsourcing.
Transparency requirements	Prompt excerpts, process note, annotated submission, declaration statement, or none with rationale.
Verification expectations	None, targeted comprehension check, viva, in-class follow-up, supervised micro-task.
Scoring rule	What the score is intended to mean under these conditions and how AI use is treated in criteria.
Equity and support provisions	Access parity, training, exemplars, accommodations, alternatives.
Data and privacy constraints	What material may not be entered into tools; approved platforms/settings.
Stability window and review trigger	Tool version or period assumed; conditions under which the task must be reviewed again.

Appendix B. Overview of potential validity threats and cross-cutting consequences

Table B1. Inference-specific validity threats in AI-inclusive assessment.

Inference	Core warrant	Illustrative AI-related validity threat
Domain	Tasks elicit performance representative of the intended construct and domain under the specified AI conditions.	**Construct underrepresentation through delegation:** GenAI enables completion of the task without the student enacting important elements of the targeted reasoning, synthesis, problem solving, or knowledge construction.
		Unacknowledged construct shift: The task increasingly elicits prompting, AI-output evaluation, or tool orchestration, while the resulting score continues to be interpreted as evidence of independent disciplinary capability.
		Underrepresentation through over-restriction: Prohibiting AI excludes tool-supported judgement, verification, or application that forms a relevant part of the contemporary academic or professional construct.
Scoring	Observed performance is evaluated according to relevant criteria, with sufficient accuracy and without material influence from factors outside the intended construct.	**Construct-irrelevant variance from access or AI fluency:** Where these are not part of the intended construct, differences in model access, prompting experience, AI literacy, or available support affect scores independently of the targeted capability.
		Marker bias concerning AI use: Disclosed or suspected AI assistance affects grading independently of the quality criteria, so otherwise comparable performances receive different scores.
		Criterion obsolescence or loss of discrimination: Rubrics reward forms of surface polish that GenAI can readily supply, compressing score distributions or weakening distinctions between stronger and weaker enactments of the intended construct.
		Detector or classification error: An unreliable or systematically biased classification of a submission as AI-generated influences its score or evaluation.
Generalisation	Scores and their interpretations remain sufficiently stable across relevant samples of tasks, raters, occasions, cohorts, and assessment conditions.	**Tool-conditioned instability:** Scores depend materially on the model, version, settings, or platform available on a particular occasion or to a particular cohort.
		Limited task or interaction sampling: Performance in a single AI-supported task or interaction is taken to represent capability across a wider range of problems, prompts, genres, or disciplinary situations.
		Rater or occasion instability: Markers differ in how they interpret acceptable AI use, disclosure, or evidence from a follow-up activity, or the same student performs differently across short verification occasions.
		Inconsistent implementation conditions: Scores depend on local variations in access, scaffolding, instructions, enforcement, or staff support that are not represented in the intended assessment universe.
Explanation	Observed performance and scores can be attributed to the targeted student capability rather than to plausible alternative causes.	**Misattribution of performance:** A polished product is attributed to the student's targeted capability even though the student cannot explain, defend, revise, or otherwise demonstrate control of the underlying reasoning.

(*Continued*)

Inference	Core warrant	Illustrative AI-related validity threat
		Conflation of authorship with competence: Unaided production is treated as sufficient evidence of understanding, or any AI assistance is treated as evidence that the student did not exercise the targeted capability. **Opaque co-production:** The submitted product provides insufficient evidence to distinguish the student's substantive judgments, verification, and decisions from contributions generated by the system. **Weak complementary attribution evidence:** A brief oral check or supervised follow-up is treated as decisive even though it elicits performance under different conditions and may also reflect oral fluency, anxiety, language background, or accessibility factors.
Extrapolation	An interpretation supported under the assessment conditions extends to the relevant target performances and contexts beyond the immediate task.	**Extrapolation from AI-supported to AI-restricted performance:** Evidence of competent AI-supported work is used to certify independent capability in contexts where equivalent tools or support are unavailable or prohibited. **Extrapolation from AI-restricted to AI-supported performance:** Performance in a secure or unaided assessment is treated as sufficient evidence of capability in professional contexts requiring responsible AI use, verification, and tool-mediated judgement. **Tool or ecosystem mismatch:** Capability demonstrated with a particular platform, scaffold, dataset, or set of permissions is assumed to transfer to target settings with different tools, constraints, risks, or responsibilities. **Weak transfer or predictive relationship:** Strong AI-supported assessment performance does not correspond to subsequent performance on related tasks, retained understanding, or competent action in relevant academic or professional settings.
Utilisation	Supported interpretations provide sufficient and appropriately weighted grounds for the specified decision or action.	**Overextension of bounded evidence:** A result that supports a limited claim about AI-supported performance is treated as sufficient for progression, certification, selection, or another broader decision. **Unsupported combination or decision rule:** AI-supported coursework, oral follow-up, supervised work, or other evidence sources are weighted or allowed to override one another without a defensible rationale. **Disproportionate integrity action:** Detector output, uncertain attribution evidence, or weak performance in a follow-up activity triggers a zero grade or misconduct process without adequate corroboration, review, or allowance for error. **Formative misuse:** AI-supported performance is interpreted as secure mastery, leading students or educators to make poorly founded decisions about feedback, support, readiness, or subsequent learning.

Table B2. Consequences and incentives as cross-cutting considerations.

Cross-cutting consideration	Illustrative AI-related concern
Learning and incentive effects	An assessment rewards polished products obtainable through delegation more strongly than engagement in learning-critical reasoning, reading, synthesis, or revision. Conversely, well-aligned conditions may encourage verification, reflection, and responsible AI-supported judgement.
Equity and access	Students differ in access to paid models, suitable devices, AI literacy, language support, accommodations, or institutional guidance. Security-oriented responses may also impose unequal burdens on particular groups.
Trust and participation	Extensive surveillance, detector use, or routine oral verification may create adversarial relationships, discourage disclosure, or alter how students participate and perform.
Workload and sustainability	Oral checks, process documentation, moderation, and individual investigations require substantial staff and student time and may be implemented inconsistently when resources are limited.
Privacy, data governance, and responsibility	Required AI use may expose personal, confidential, copyrighted, or professionally sensitive material, while monitoring practices may collect intrusive process data. Responsibility for erroneous or inappropriate AI-supported outputs may also remain unclear.